Session Objectives

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College of Osteopathic Medicine

At the conclusion of the session/course, the student should be able to:

1. Discuss the various mechanisms of acquired immune deficiency.
2. Gauge the type and severity of immune deficiency in a patient presenting with opportunistic infection.
3. Differentiate opportunistic pathogens that may present in patients with immune deficiency.
4. Diagnose, treat and manage and patient with severe immune deficiency secondary to AIDS.
5. Identify the risk factors for immune reconstitution inflammatory syndrome.

Source: Course Syllabus

Types of Immune Deficiencies

Deficiency	Examples	Organisms
Extremes of age	Birth, elderly	Zoster
Broad-spectrum abx	Sepsis Resistant organisms rx	Pseudomonas Candida
Foreign bodies	Central venous catheter Urinary catheter Joint replacement Heart valve replacement Pacemaker	*Staphylococcus epidermidis*, *S. aureus*, enterococci, *Corynebacterium* *Candida*
Trauma	Penetrating injury Surgery Burns	*S. epidermidis*, *S. aureus*, *Pseudomonas aeruginosa*
Chronic illness	Diabetes Chronic renal failure COPD, cystic fibrosis	*Staphylococcus aureus*, *S. epidermidis*, enteric Gram negatives, *Pseudomonas*, *Streptococcus pneumoniae* *Haemophilus influenzae*, NTMB
Complement	Properdin deficiency SLE Eculizumab	*Neisseria* infections

Question

Increased risk of infection with which of the following organisms occurs after autosplenectomy?

1. *Babesia*
2. *Streptococcus pneumoniae*
3. *Capnocytophagia*
4. *Staphylococcus epidermidis*

Types of Immune Deficiencies

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Deficiency	Examples	Organisms
Alpha-4 (α4) integrin	Natalizumab	PML
Humoral	Bruton's Multiple myeloma CLL Small lymphocyte lymphoma Splenectomy Common variable immunodeficiency	*Haemophilus influenzae, Neisseria meningiditis, Streptococcus pneumoniae, Capnocytophagia* Babesiosis
Granulocytopenia	Hematopoietic cell transplantation Myelosuppressive chemotherapy Acute leukemia	Gram-negative enterics, *Pseudomonas, Staphylococcus aureus, S. epidermidis*, Viridans streptococci *Candida, Aspergillus, Trichosporon, Scedosporium, Fusarium,* mucormycosis
Cellular immunity	HIV infection Lymphoreticular malignancies Corticosteroids, cyclosporine, tacrolimus Solid organ transplantation TNF inhibitors Idiopathic CD4 lymphopenia	*Listeria, Legionella, Salmonella, Mycobacterium* HSV, VZV, CMV, EBV, HBV, HCV *Cryptococcus, Coccidioides, Histoplasma, Pneumocystis* *Toxoplasma*

Question

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Risk of infection by Aspergillus sp. Is increased in which phase of allogeneic hematopoietic stem cell transplant?

1. Pretransplantation
2. Pre-engraftment (1 – 21 days)
3. Post-engraftment (3 weeks – 3 months)
4. > 3 months

Hematopoietic Cell Transplantation

- Early post-transplant (pre-engraftment) 1-21 days
  - Severe neutropenia up to 21 days
    - Staphylococci, gram-negative, HSV, RSV, *Candida*, *Aspergillus*
- 3 weeks – 3 months (post-engraftment)
  - CMV, adenovirus, *Aspergillus*, *Candida*, *Pneumocystis*
- >3 months
  - VZV, *Aspergillus*, CMV

Solid Organ Transplantation

- Immediate postoperative
  - Infections involving transplanted organ
- 2-4 weeks posttransplant
  - Wound, IV catheter, UTI
  - Transplanted infections
- 1-6 months posttransplant
  - HSV, VZV, CMV, *Candida, Aspergillus, Cryptococcus, Pneumocystis, Listeria monocytogenes, Nocardia, Toxoplasma*

Risk of posttransplant CMV infection (1-6 months)

High risk—CMV Ab (−) recipient, CMV Ab (+) donor
Intermediate risk—CMV Ab (+) recipient, CMV Ab (+) or Ab (−) donor
Lowest risk—CMV Ab (−) recipient, CMV Ab (−) donor

Question

Which of the following increases risk of infection in patients with absolute neutropenia?

1. Duration >7 days
2. Duration <7 days
3. Granulocyte count >500/ml
4. Granulocyte count >1000/ml

Bobby

He explains that 8 years ago he broke up with his partner of 4 years and returned to using injection drugs, mostly heroin. He stopped taking ART then and over the last year has been feeling unwell. He has lost 35 pounds and was hospitalized 8 months ago for 5 days with pneumonia. He has not otherwise seen a physician.

Bobby

- The differential diagnosis in those with severe combined immune deficiency is broad. For those with advanced HIV/AIDS, critically ill and ground glass, pulmonary opacities on CxR:
  - *Pneumocystitis jirovecii* pneumonia, tuberculosis, *S. pneumoniae*, *S. aureus* (including MRSA), *P. aeruginosa*, influenza, HSV, CMV, VZV, *Histoplasma*, *Coccidioides*, pulmonary lymphoma with lymphangitic spread, acute pulmonary edema, ARDS

Question

Which of the following is most common opportunistic infection in persons living with AIDS not on antiretroviral therapy?

1. Cytomegalovirus retinitis
2. *Mycobacterium avium/intracellulare*
3. Cryptococcal meningitis
4. *Pneumocystis jirovecii* pneumonia

Bobby

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TMP/SMZ IV, prednisone, vancomycin, meropenem and fluconazole are begun.

Labs: CD4 3, HIV-1 PCR >1,000,000/ml, LDH 949 (elevated), blood cultures, beta-D-glucan and CMV PCR neg, pancytopenia, elevated transamininases, acute renal insufficiency.

Sputum by tracheal suction was *Pneumocystis* antigen positive.

Bobby

CMV PCR of an ulcer is positive. Ganciclovir IV is begun, and odynophagia improves. Three days later on hospital day 12, he has new chills associated with diaphoresis and tachycardia. He has no fever. He has also noted mild axillary submental and inguinal adenopathy. Pancytopenia persists. Alkaline phosphatase increases to 4x normal. Mycobacterial sputum and blood cultures obtained on admission are reported from the lab to be growing an organism.

What procedure is performed?

Pause

Question

A 32-year-old man living with AIDS, CD4 2/ml, presents fever, chills, adenopathy and severe anemia. Bone marrow biopsy shows organisms that stain with Ziehl-Neelsen. Treatment of this organism should start with which of the following?

1. Ethambutol and rifampin
2. Azithromycin and ethambutol
3. Clarithromycin and rifampin
4. Isoniazid and rifampin

Bobby

Bone marrow biopsy shows organisms on Ziehl-Neelsen stain and sputum PCR is positive for MAI. Azithromycin and ethambutol are begun. TMP/SMZ, prednisone, valganciclovir are continued. On hospital day 16, he complains of a vague headache that may have been present for several days, but today is severe enough to be bothersome and he notes mild-moderate neck stiffness.

What procedure is performed?

Pause

Meningitis in AIDS

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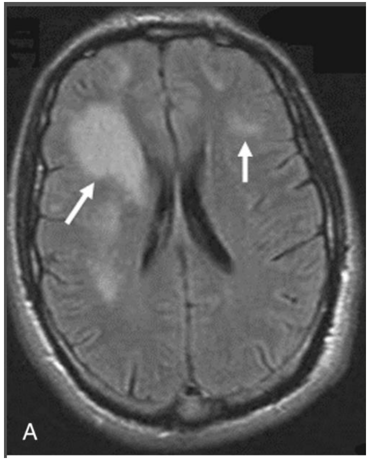
- DDx: Cryptococcus neoformans, *Histoplasma capsulatum*, *Coccidioides immitis*, *M. tuberculosis*, syphilis, *Listeria*, Lymphomatous, aseptic
- Fever, headache, photophobia, neck stiffness
  C. neoformans may be asymptomatic
  Altered mental status may indicate increased intracranial pressure, risk for uncal herniation
- Dx: Lumbar puncture

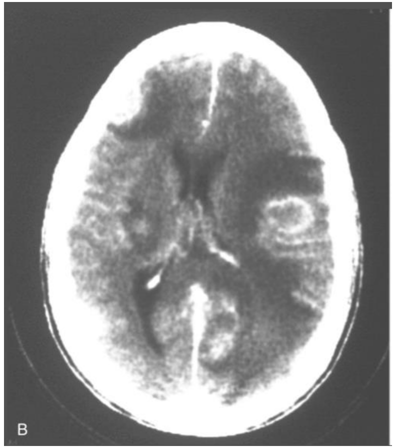
Question

The organism that most commonly causes meningitis in those living with advanced AIDS is associated with which of the following?

1. Travel to the southwest U.S.
2. Exposure to arboreal species
3. Acute retinal necrosis
4. Asymptomatic infection

Other Cerebral Opportunistic Infections





A. Progressive Multifocal Leukoencephalopathy (PML)

- JC polyomavirus (nonenveloped, circular dsDNA)
- Immune competent: 90% are infected (asymptomatic)
  Immune suppressed, (CD4 <50, natalizumab): Uni- or multifocal cerebral demyelination
- Subacute cognitive impairment, focal motor deficits, and seizures, MRI brain shows demyelinating lesions
- MRI: White matter demyelinating lesions
- DDx: AIDS dementia complex, CMV encephalitis, infarction
- Dx: CSF JC virus NAT
- Treatment: ART

B. *Toxoplasma gondii*

- Protozoan parasitic infection
- Ingestion of tissue cysts in raw or undercooked meat, mother to fetus
- Immune competent: asymptomatic, seizures from calcified cerebral cysts, chorioretinitis
  Immune suppressed (CD4 <50): chorioretinitis,
- Fever, headache, confusion, motor defects, and seizures
- CT or MRI: space-occupying gray matter lesion
- DDx: lymphoma
- Dx: CSF Toxoplasma PCR, serology, pathology
- Treatment: sulfadiazine-pyrimethamine or clindamycin-pyrimethamine

Bobby

Lumbar puncture shows glucose of 10 mg/dl, protein 220 mg/dl, WBC 1200/ml with 80% mononuclear cells. India ink is positive and cryptococcal antigen titer is 1:64. Opening pressure was 22 mmH_2O.

Liposomal amphotericin B and flucytosine are begun with the plan to switch to oral fluconazole in 2 weeks. His headache and meningismus improve. Discharge planning is begun and his medications at home will also include TMP/SMZ, valganciclovir, azithromycin and ethambutol.

When Bobby asks why is not going home on ART, what do you tell him?

Question

Which of the following organisms is associated with immune reconstitution inflammatory syndrome when ART is begun when the CD4 is 200?

1. Mycobacterium avium adenitis
2. Tuberculosis
3. Cytomegalovirus retinitis
4. *Pneumocystis jirovecii* pneumonitis

Immune Reconstitution Inflammatory Syndrome (IRIS)

- Initiation of ART results in inflammatory reaction that results in tissue injury
  - Unmasking IRIS: Undiagnosed, occult infection
  - Paradoxical IRIS: Residual antigenic stimulation
- CD4 prior to ART usually <100 (TB <200)
- Occurs in up to 13% one week to 3-4 months after ART initiation

- *Manifestations and treatment dependent on organ and organism*
  - *Mycobacterium tuberculosis*
  - *Mycobacterium avium complex*
  - Cytomegalovirus
  - *Cryptococcus neoformans*
  - *Pneumocystis jirovecii*
  - Herpes simplex virus
  - Hepatitis B virus
  - Human herpes virus 8 (associated with Kaposi sarcoma)
  - PML
- Prevention: treat known infection 2-4 weeks prior to initiating ART